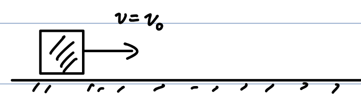
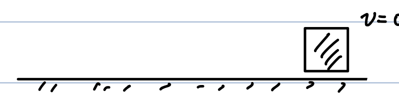
Thermodynamics
↓
Heat - related

Last time we encounter "heat":





kinetic friction : μN

Total mechanical energy E_{mec}: $\frac{1}{2}mv^2 \longrightarrow 0$

Total energy conserved? Yes

$$E = E_{mec} + E_{thermo} + \cdots$$

Temperature

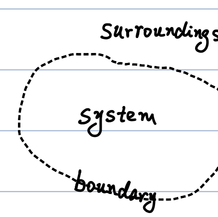
(operational definition) number on a thermometer.

How to measure?

- contact (energy transfer)
- wait (relaxation)
- read (convert $T \rightarrow \#$)

Thermodynamic system (abbr. "system")

A macroscopic body separate from its surroundings.



e.g. a moving box,
boiling water,
a can of gas ...

- Open system: exchange energy & matter with surroundings
- Closed system: exchange energy but not matter
- Isolated system: No energy or matter exchange

Thermal equilibrium

When macroscopic properties of a system do NOT change with time
(e.g. temperature, volume, ⋯)

Note: microscopic properties still changing (thermal equilibrium is "dynamic")
e.g. positions of H_2O molecules in still water

The zeroth law of thermodynamics

1931, Ralph H. Fowler

If two systems are both in thermal equilibrium with a third system, then the two systems are in thermal equilibrium with each other.

$$A \Leftrightarrow C,\ C \Leftrightarrow B \quad \Longrightarrow \quad A \Leftrightarrow C,\ C \Leftrightarrow B,\ A \Leftrightarrow B$$

"zeroth": $\begin{cases} \text{established after other laws} \\ \text{logically before other laws ("temperature" is fundamental)} \end{cases}$

Temperature

Measure of the tendency of a system to spontaneously change state (out of equilibrium) when put in contact with another system.

In equilibrium $\Longleftrightarrow$ Same temperature

Zeroth law: $\begin{cases} T_A = T_C \\ T_B = T_C \end{cases} \Rightarrow T_A = T_B$

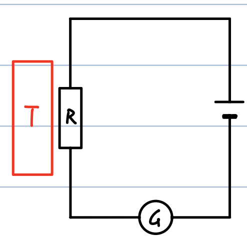
Temperature measured by thermometer:

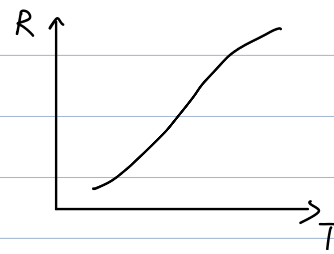
- contact (two systems with different T's)
- relax (evolves towards new equilibrium)
- measure (how?)

- Mercury thermometer :
  - Volume changes with temperature.

- Other thermometers :
  - Length of solid
  - Electric resistance of conductor
  - Pressure/Volume of gas at constant volume/pressure
  - ...

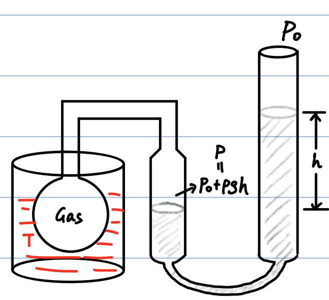
Thermometer based on resistence

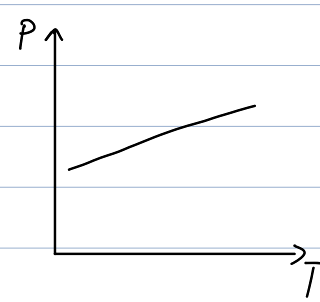




R(T) relation used to calibrate the thermometer

Gas thermometer

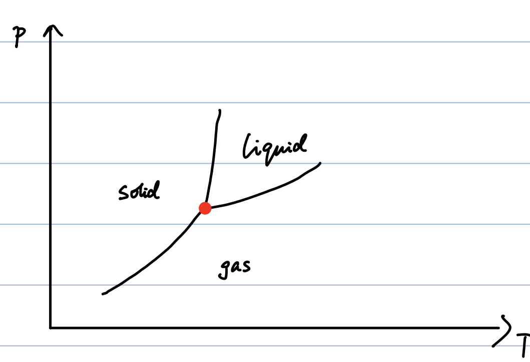




P(T) for calibrating the thermometer

How to calibrate?

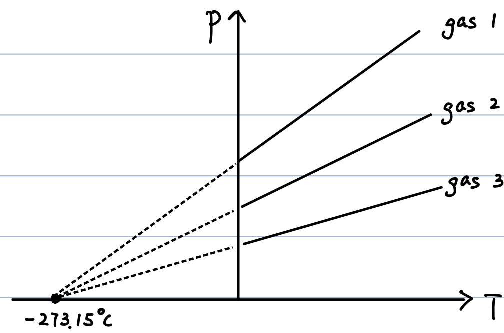
pick a common (easily-reproducible) phenomena: triple point of water



$T_3 \equiv 273.16\ K$

$\equiv 0.01\ ^\circ C$

Absolute zero



Celsius $\longleftrightarrow$ Kelvin :

$$\boxed{T^{(C)} = T^{(K)} - 273.15}$$

i.e. $T = 0\ K \Longleftrightarrow T = -273.15°C$

Fahrenheit $\leftrightarrow$ Celsius :

$$T^{(F)} = \frac{9}{5} T^{(C)} + 32$$

Absolute temperature (kelvin) scale

$1\,K \equiv \frac{1}{273.16} T_3$ (old definition)

T_3: temperature of triple point of water

$k_B = 1.380649 \times 10^{-23}$ J/K (new definition)

(1 K change of temperature $\leftrightarrow$ thermal energy change of 1.380649×10^{-23} J)

The third law of thermodynamics

1912, Walther Nernst

"It is impossible for any procedure to lead to the isotherm $T=0$ in a finite number of steps."

Note: Various forms of the third law exist.

	Kelvin	Celsius	Fahrenheit
Boiling of H_2O	373.15 K	100°c	212 °F
Freezing of H_2O	273.15 K	0°c	32 °F
Boiling of N_2	77.36 K	-195.79°C	-320.42 °F
Absolute zero	0 K	-273.15°C	-459.67 °F

Thermal expansion

- Linear expansion $\qquad \Delta L = \alpha L_0 \Delta T$
- Volume expansion $\qquad \Delta V = \beta V_0 \Delta T$

α, β : expansion coefficients, which depend on

- distance between molecules
- size of molecules
- range of interactions

Relation between α & β ?

$$V = L^3 = (L_0 + \Delta L)^3$$
$$= L_0^3 (1 + \alpha \Delta T)^3$$
$$\approx \underbrace{L_0^3}_{V_0} [1 + 3\alpha \Delta T + O(\Delta T)^2]$$

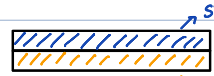
$$\Rightarrow \quad \boxed{\beta = 3\alpha}$$

Examples of thermal expansion

- Mercury thermometer (previous)
- Thermostat with bimetallic stripe

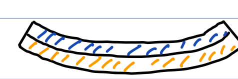
Room temperature

(iron + carbon)
steel $\quad \alpha = 11 \times 10^{-6} \; (^\circ C)^{-1}$



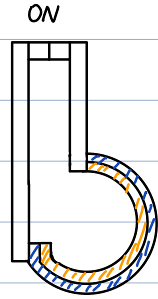
brass $\quad \alpha = 19 \times 10^{-6} \; (^\circ C)^{-1}$
(copper + zinc)

High temperature



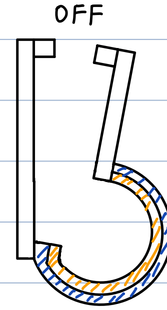
$\downarrow$

ON



$25^\circ C$

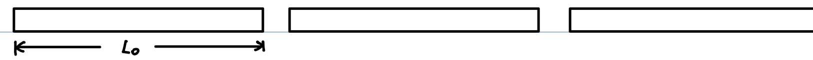
OFF



$30^\circ C$

- railroad track

traditional:



$L_0 = 30\,m$ at $0°c$

Q: L at $40°c$?

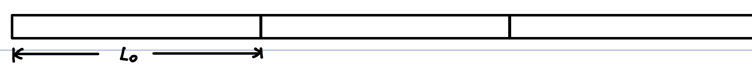
Steel: $\alpha = 11 \times 10^{-6}\ (°c)^{-1}$

$\Rightarrow \Delta L = \alpha L_0 \Delta T$

$= 11 \times 10^{-6} \times 30 \times 40$

$= 0.0132\ m$

high-speed rail:



Cost? tensile stress at $40°c$:

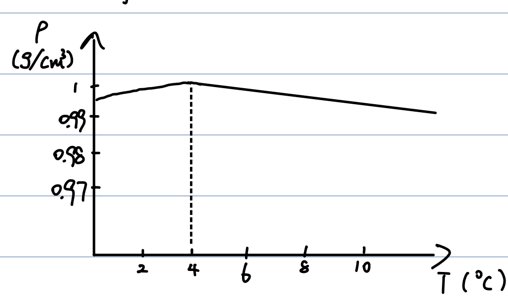
$$\frac{F}{A} = \underbrace{E}_{\text{Young's modulus}} \frac{\Delta L}{L} = (20 \times 10^{10}\ N/m^2) \times \frac{0.0132m}{30\,m} = 8.8 \times 10^{7}\ N/m^2$$

With $A = 10\,cm^2 = 10^{-3}\,m^2 \Rightarrow F = 8.8 \times 10^{4}\ N \sim 10^{4}\ kg$

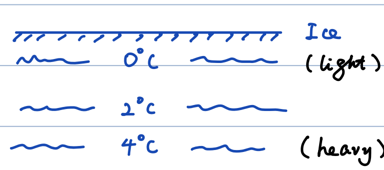
$(1kg \sim 9.8\,N)$

- Exception - Water

$\beta < 0$ for $0°c < T < 4°c$: thermal contraction!



Freezing:



- Ideal gas

$$\boxed{pV = nRT}$$

$n \equiv \frac{N}{N_A}$ $\qquad N_A = 6.02 \times 10^{23}\ \mathrm{mol}^{-1}$ Avogadro's number

$R = 8.31\ \mathrm{J \cdot mol^{-1} \cdot K^{-1}}$ $\qquad$ gas constant

Another form : $\boxed{pV} = \frac{N}{N_A} RT \boxed{= N k_B T}$ $\qquad$ i.e. $R = N_A k_B$

$$\Delta V = \beta V_0 \Delta T \longrightarrow \beta = \frac{1}{V_0}\frac{\Delta V}{\Delta T}$$
$$= \left(\frac{1}{V}\frac{dV}{dT}\right)_p$$
$$= \left(\frac{d \ln V}{dT}\right)_p$$
$$= \left(\frac{d[\ln T + \ln(nR/p)]}{dT}\right)_p$$
$$= \frac{1}{T}$$

Ideal gas at $0^\circ\mathrm{C}$: $\beta = \frac{1}{273.15} \approx 0.00366\ (^\circ\mathrm{C})^{-1}$

- Real gas

Van der Waals equation of state :

$$\boxed{\left(p + \frac{aN^2}{V^2}\right)(V - Nb) = N k_B T}$$

b : volume of a molecule $\Rightarrow V - Nb$ corrected volume at high pressure

a : from interactions of molecules